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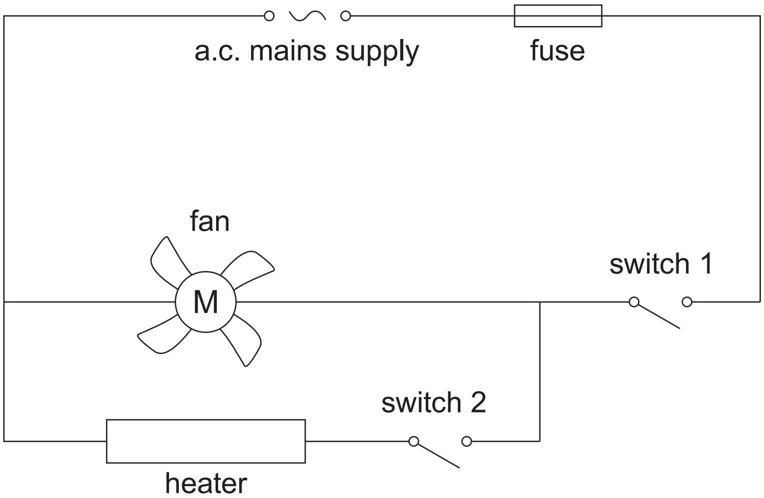
2. The circuit diagram below shows a design for a hairdryer that can blow either cold air or hot air.

The switches in the diagram are in the open position.



(a) Complete the following sentences by underlining the correct phrase in the brackets. [2]

When switch 1 is closed and switch 2 is open the hairdryer blows cold air because

(**only the fan is on** / **only the heater is on** / **the heater and fan are both on**).

The hairdryer blows hot air when the heater and fan are both on. This happens when

(**both switches are open** / **only switch 2 is closed** / **both switches are closed**).

(b) The hairdryer is used to blow hot air.

The current supplied is **6 A** and the mains voltage is 230 V.

(i) Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the hairdryer. [2]

power = ... W

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(ii) The following fuses are available.

Tick (✓) **the box** to show the correct fuse that should be used for the hairdryer. [1]

☐ ☐ ☐ ☐

(iii) Complete the following sentence by underlining the correct phrase in the brackets. [2]

Fuses switch off circuits if the
(current is too high / voltage is too high / power is too low)
to prevent **(electric shocks / wires overheating / power cuts)**.

(c) A **different hairdryer** is now used. The power of this hairdryer is **1.5 kW**.

(i) Use the equation:

$$\text{units used (kWh)} = \text{power (kW)} \times \text{time (h)}$$

to calculate the units used (kWh) by this hairdryer in **2 hours**. [2]

units used = ... kWh

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(ii) The cost per unit of electrical energy is 30 p.
Use your answer to part (c)(i) and the equation:

$$\text{cost} = \text{units used} \times \text{cost per unit}$$

to calculate the cost of using this hairdryer for 2 hours. [2]

cost = ... p

(iii) State the cost of using this hairdryer for **double** the time. [1]

cost = ... p

(iv) Tick (✓) **the box** that correctly shows the power of the 1.5 kW hairdryer in watts (W). [1]

- ☐ 1 500 000 W
- ☐ 1 500 W
- ☐ 1.5 W
- ☐ 0.015 W

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